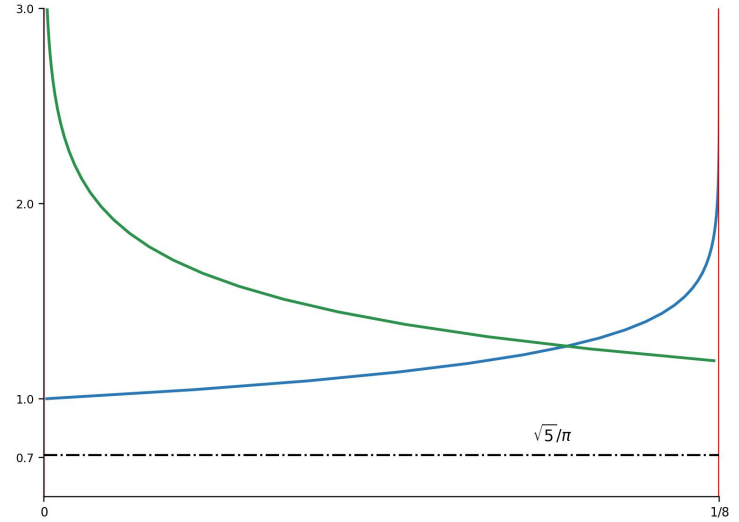
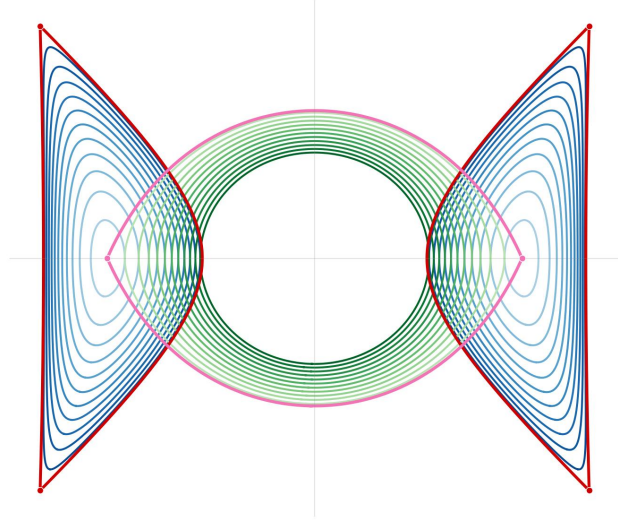




Curve $\alpha = 2H(p, q) = p^2 + q^2 + q^3 - 3p^2q + \frac{p^4 - 6p^2q^2 + q^4}{4}, \quad \rho = (2H)_p = p(p^2 + 2 - 6q - 3q^2).$

Period form $\omega = \frac{2dq}{\rho} = \frac{dq}{H_p}.$

Real period $T_1(z) = \frac{1}{2\pi} \oint_{\gamma_z^{(1)}} \omega, \quad T_1(0) = 1, \quad \gamma_z^{(1)} \subset (p, q) \text{ around } (0, 0).$

Complex period $T_2(z) = \frac{\sqrt{5}}{4\pi} \oint_{\gamma_z^{(2)}} i\omega, \quad T_2(1/4) = 1, \quad \gamma_z^{(2)} \subset (ip, q) \text{ around } (0, -1).$

Abel's identity $\frac{z(z+6)(4z-1)(8z-1)}{17z-3} (T_1(z)T_2'(z) - T_2(z)T_1'(z)) = \frac{\sqrt{5}}{\pi}.$

Annihilator $A_\alpha = P_0 + P_1 \partial_\alpha + P_2 \partial_\alpha^2.$

$$P_0 = 27 - 261\alpha + 645\alpha^2 + 408\alpha^3.$$

$$P_1 = -18 + 426\alpha - 2827\alpha^2 + 5736\alpha^3 + 1632\alpha^4.$$

$$P_2 = \alpha(\alpha+6)(4\alpha-1)(8\alpha-1)(17\alpha-3).$$

Factorization $P_1 = P_2' - \frac{34}{17\alpha-3} P_2, \quad A_\alpha = (17\alpha-3)^2 \partial_\alpha \left(\frac{P_2}{(17\alpha-3)^2} \partial_\alpha \right) + P_0.$

Certificate $\Xi = \frac{V(\alpha, p, q)}{\rho^3}, \quad A_\alpha \circ \omega = d\Xi, \quad V = \sum_{j=0}^9 u_j(\alpha) q^j + p^2 \sum_{j=0}^7 v_j(\alpha) q^j.$

Numerator: q^j coefficients

$$u_0 = -6\alpha(136\alpha^3 + 411\alpha^2 - 211\alpha + 14)$$

$$u_1 = -2(1224\alpha^4 - 6765\alpha^3 + 3412\alpha^2 - 381\alpha + 18)$$

$$u_2 = -2448\alpha^4 + 4410\alpha^3 + 6337\alpha^2 - 2811\alpha + 210$$

$$u_3 = -816\alpha^4 - 31170\alpha^3 + 7331\alpha^2 + 1719\alpha - 66$$

$$u_4 = -2(12240\alpha^3 + 9535\alpha^2 - 5142\alpha + 363)$$

$$u_5 = -2(2448\alpha^3 - 6185\alpha^2 + 2256\alpha + 39)$$

$$u_6 = 14(3842\alpha^2 - 1473\alpha + 60)$$

$$u_7 = 2(20162\alpha^2 - 7665\alpha + 348)$$

$$u_8 = 36(340\alpha^2 - 129\alpha + 6)$$

$$u_9 = 4(340\alpha^2 - 129\alpha + 6)$$

Numerator: $p^2 q^j$ coefficients

$$v_0 = -816\alpha^4 - 650\alpha^3 + 3475\alpha^2 - 1089\alpha + 102$$

$$v_1 = -3(272\alpha^4 - 2322\alpha^3 + 8515\alpha^2 - 2733\alpha + 258)$$

$$v_2 = -2(816\alpha^3 - 16195\alpha^2 + 5646\alpha - 543)$$

$$v_3 = -2(11152\alpha^3 - 37113\alpha^2 + 11910\alpha - 1125)$$

$$v_4 = -10(1632\alpha^3 + 6286\alpha^2 - 2343\alpha + 228)$$

$$v_5 = -2(1632\alpha^3 + 61502\alpha^2 - 21579\alpha + 2076)$$

$$v_6 = -28(1972\alpha^2 - 687\alpha + 66)$$

$$v_7 = -4(1972\alpha^2 - 687\alpha + 66)$$

Series

$$T_1(8x) = 1 + 12x + 372x^2 + 15120x^3 + 706020x^4 + 35692272x^5 + \dots,$$

$$T_2(1/4 - 100x) = 1 + 108x + 24420x^2 + 6819120x^3 + 2096600100x^4 + 681263917488x^5 + \dots.$$